



МИНОБРНАУКИ РОССИИ
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«МИРЭА - Российский технологический университет»
РТУ МИРЭА

Практическое занятие № 7

Тема занятия: Разработка СУБД Clickhouse

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Ефремов ЭФБО-10-23 Практическая 7

7.1 Разработка структуры базы данных

Задание: Установить и настроить ClickHouse. Создать БД в ClickHouse со своей фамилией, добавить свою фамилию и имя, где только возможно, кроме названий таблиц.

Код установки (в Docker):

```
docker pull clickhouse/clickhouse-server
docker run -d --name some-clickhouse-server --ulimit
nofile=262144:262144 clickhouse/clickhouse-server
docker exec -it some-clickhouse-server clickhouse-client
```

```
Amirea_lab/5 sem/bd / ? main * !? > docker run -d --name some-clickhouse-server --ulimit nofile=262144:262144 clickhouse/clickhouse-server @ 11:26
7ad9c928729f9f30848f213a95b33363b02f08e880810ab0bbe2293c9cf41226
Amirea_lab/5 sem/bd / ? main * !? > docker exec -it some-clickhouse-server clickhouse-client @ 11:29
ClickHouse client version 25.11.2.24 (official build).
Connecting to localhost:9000 as user default.
Connected to ClickHouse server version 25.11.2.

Warnings:
* Delay accounting is not enabled, OSIOWaitMicroseconds will not be gathered. You can enable it using 'sudo sh -c 'echo 1 > /proc/sys/kernel/task_delayacct'' or by using sysctl.
* Linux transparent hugepages are set to "always". Check /sys/kernel/mm/transparent_hugepage/enabled
* Linux threads max count is too low. Check /proc/sys/kernel/threads-max

7ad9c928729f :) \dt
```

Создание БД:

```
CREATE DATABASE yefremov_db;
USE yefremov_db;
```

```
7ad9c928729f :) CREATE DATABASE yefremov_db;
USE yefremov_db;

CREATE DATABASE yefremov_db

Query id: c0391af3-90a6-43f0-a3cc-839e886daa5d

Ok.

0 rows in set. Elapsed: 0.021 sec.

USE yefremov_db

Query id: f7e204c1-e41b-4b3a-835f-d62cca851298

Ok.

0 rows in set. Elapsed: 0.002 sec.

7ad9c928729f :) █
```

Скриншот: Скриншот выполнения CREATE DATABASE и SHOW DATABASES с видимой yefremov_db.

Урок 1: Введение в ClickHouse

Задание: Знакомство с ClickHouse, создание простой таблицы.

Код:

```
CREATE TABLE yefremov_intro (  
    id UInt32,  
    name String COMMENT 'Yefremov Aleksey'  
) ENGINE = MergeTree() ORDER BY id;  
  
INSERT INTO yefremov_intro VALUES (1, 'Yefremov Aleksey');  
SELECT * FROM yefremov_intro;
```

```
7ad9c928729f :) CREATE TABLE yefremov_intro (  
    id UInt32,  
    name String COMMENT 'Yefremov Aleksey'  
) ENGINE = MergeTree() ORDER BY id;
```

```
INSERT INTO yefremov_intro VALUES (1, 'Yefremov Aleksey');  
SELECT * FROM yefremov_intro;
```

```
CREATE TABLE yefremov_intro  
(  
    `id` UInt32,  
    `name` String COMMENT 'Yefremov Aleksey'  
)  
ENGINE = MergeTree  
ORDER BY id
```

Query id: ae47946b-dae4-4450-9376-837d0cb72713

Ok.

0 rows in set. Elapsed: 0.020 sec.

```
INSERT INTO yefremov_intro FORMAT Values
```

Query id: 93c62ef5-9fae-4b23-bf24-6bf0e24e036d

Ok.

1 row in set. Elapsed: 0.007 sec.

```
SELECT *  
FROM yefremov_intro
```

Query id: 5fcd101-9036-4da2-905d-d5c057e3d61f

	id	name
1.	1	Yefremov Aleksey

1 row in set. Elapsed: 0.003 sec.

7ad9c928729f :) █

Урок 2: Типы данных и движки в ClickHouse

Задание: Создать таблицу с разными типами данных, использовать MergeTree.

Код:

```
CREATE TABLE yefremov_types (
    id UInt64,
    name String,
    birth_date Date,
    score Float32 COMMENT 'Yefremov Aleksey score'
) ENGINE = MergeTree() ORDER BY id;

INSERT INTO yefremov_types VALUES (1, 'Yefremov Aleksey', '2000-01-01', 95.5);

SELECT * FROM yefremov_types;
```

```
7ad9c928729f :) CREATE TABLE yefremov_types (
    id UInt64,
    name String,
    birth_date Date,
    score Float32 COMMENT 'Yefremov Aleksey score'
) ENGINE = MergeTree() ORDER BY id;

INSERT INTO yefremov_types VALUES (1, 'Yefremov Aleksey', '2000-01-01', 95.5);
SELECT * FROM yefremov_types;

CREATE TABLE yefremov_types
(
    `id` UInt64,
    `name` String,
    `birth_date` Date,
    `score` Float32 COMMENT 'Yefremov Aleksey score'
)
ENGINE = MergeTree
ORDER BY id

Query id: 313998f5-a1b1-4f9a-a2ce-daba856eec61

Ok.

0 rows in set. Elapsed: 0.020 sec.

INSERT INTO yefremov_types FORMAT Values

Query id: d13fd918-e5bf-4939-a910-d6816eb2113b

Ok.

1 row in set. Elapsed: 0.009 sec.

SELECT *
FROM yefremov_types

Query id: 657fb872-85e0-48f3-ae68-03d688f41aed

1. 

| id | name             | birth_date | score |
|----|------------------|------------|-------|
| 1  | Yefremov Aleksey | 2000-01-01 | 95.5  |



1 row in set. Elapsed: 0.004 sec.

7ad9c928729f :) █
```

Урок 3: Работа с данными (INSERT, SELECT)

Задание: Вставка данных, простые запросы.

Код:

```
CREATE TABLE yefremov_data (  
    id UInt32,  
    event String  
) ENGINE = MergeTree() ORDER BY id;  
  
INSERT INTO yefremov_data VALUES (1, 'Yefremov Aleksey event'),  
(2, 'Another event');  
SELECT count() FROM yefremov_data WHERE event LIKE '%Yefremov%';
```

```
7ad9c928729f :) CREATE TABLE yefremov_data (  
    id UInt32,  
    event String  
) ENGINE = MergeTree() ORDER BY id;  
  
INSERT INTO yefremov_data VALUES (1, 'Yefremov Aleksey event'), (2, 'Another event');  
SELECT count() FROM yefremov_data WHERE event LIKE '%Yefremov%';
```

```
CREATE TABLE yefremov_data  
(  
    `id` UInt32,  
    `event` String  
)  
ENGINE = MergeTree  
ORDER BY id
```

Query id: 36dc6e22-1ef6-43fc-8cfe-0c8c64191151

Ok.

0 rows in set. Elapsed: 0.029 sec.

```
INSERT INTO yefremov_data FORMAT Values
```

Query id: e3af0ace-7049-4239-a56c-143bb53ff68b

Ok.

2 rows in set. Elapsed: 0.008 sec.

```
SELECT count()  
FROM yefremov_data  
WHERE event LIKE '%Yefremov%'
```

Query id: e248feac-20d4-48b7-ad0d-f94174b5ecfd

```
1.  ┌count()┐  
    └───┴───┘  
      1
```

1 row in set. Elapsed: 0.006 sec.

7ad9c928729f :) █

Урок 4: Функции и агрегаты

Задание: Использовать функции (upper, count, avg).

Код:

```
SELECT upper('yefremov aleksey') AS upper_name;  
SELECT count(), avg(score) FROM yefremov_types;
```

```
7ad9c928729f :) SELECT upper('yefremov aleksey') AS upper_name;  
SELECT count(), avg(score) FROM yefremov_types;
```

```
SELECT upper('yefremov aleksey') AS upper_name
```

Query id: 74ce5dc5-c167-4342-b938-da2802a9f423

upper_name
YEFREMOV ALEKSEY

1 row in set. Elapsed: 0.003 sec.

```
SELECT  
    count(),  
    avg(score)  
FROM yefremov_types
```

Query id: 91202455-133b-414e-9ea3-d7a1f3331dca

count()	avg(score)
1	95.5

1 row in set. Elapsed: 0.006 sec.

7ad9c928729f :) 1;9u

Урок 5: JOIN и подзапросы

Задание: Создать вторую таблицу, JOIN.

Код:

```
CREATE TABLE yefremov_join1 (  
    id UInt32,  
    value String  
) ENGINE = MergeTree() ORDER BY id;  
  
INSERT INTO yefremov_join1 VALUES (1, 'Yefremov value');  
  
CREATE TABLE yefremov_join2 (  
    id UInt32,  
    detail String  
) ENGINE = MergeTree() ORDER BY id;  
  
INSERT INTO yefremov_join2 VALUES (1, 'Aleksey detail');  
  
SELECT j1.value, j2.detail  
FROM yefremov_join1 j1  
JOIN yefremov_join2 j2 ON j1.id = j2.id;
```

```

7ad9c928729f :) CREATE TABLE yefremov_join1 (
    id UInt32,
    value String
) ENGINE = MergeTree() ORDER BY id;

INSERT INTO yefremov_join1 VALUES (1, 'Yefremov value');

CREATE TABLE yefremov_join2 (
    id UInt32,
    detail String
) ENGINE = MergeTree() ORDER BY id;

INSERT INTO yefremov_join2 VALUES (1, 'Aleksey detail');

SELECT j1.value, j2.detail
FROM yefremov_join1 j1
JOIN yefremov_join2 j2 ON j1.id = j2.id;

CREATE TABLE yefremov_join1
(
    `id` UInt32,
    `value` String
)
ENGINE = MergeTree
ORDER BY id

```

Query id: c5a5bd22-396d-413b-8eb5-5eae0cae1ff5

Ok.

0 rows in set. Elapsed: 0.028 sec.

```
INSERT INTO yefremov_join1 FORMAT Values
```

Query id: 10994289-8702-494e-8450-061387cf7160

Ok.

1 row in set. Elapsed: 0.007 sec.

```

CREATE TABLE yefremov_join2
(
    `id` UInt32,
    `detail` String
)
ENGINE = MergeTree
ORDER BY id

```

Query id: 534dc6f3-2f56-400f-b5db-473bfe8c11be

Ok.

0 rows in set. Elapsed: 0.014 sec.

```
INSERT INTO yefremov_join2 FORMAT Values
```

Query id: f3af5b68-b40c-4808-8510-ae68ace23b34

Ok.

1 row in set. Elapsed: 0.007 sec.

```

SELECT
    j1.value,
    j2.detail
FROM yefremov_join1 AS j1
INNER JOIN yefremov_join2 AS j2 ON j1.id = j2.id

```

Query id: d37578a9-66fd-4eed-9dac-821c2557e895

	value	detail
1.	Yefremov value	Aleksey detail

1 row in set. Elapsed: 0.005 sec.

7.2 Работа с данными в СУБД NoSQL

Задание: Выполнить задания уроков 6-10, сделать скриншоты так, чтобы на них была видна ваша фамилия.

Урок 6: Оптимизация запросов в ClickHouse: индексы, EXPLAIN и лучшие практики

Задание: Создать таблицу с MergeTree, ORDER BY, выполнить запросы с фильтрами, EXPLAIN.

Код:

```
CREATE TABLE yefremov_access_logs (  
    event_date Date,  
    user_id UInt32,  
    timestamp DateTime,  
    action String COMMENT 'Yefremov Aleksey action'  
) ENGINE = MergeTree()  
ORDER BY (event_date, user_id);  
  
INSERT INTO yefremov_access_logs VALUES  
(  
'2025-06-01', 101, '2025-06-01 10:00:00', 'Yefremov login'),  
(  
'2025-06-02', 102, '2025-06-02 11:00:00', 'Aleksey view');  
  
SELECT count()  
FROM yefremov_access_logs  
WHERE event_date >= '2025-06-01' AND event_date < '2025-07-01'  
AND user_id IN (101, 102);  
  
EXPLAIN ESTIMATE SELECT * FROM yefremov_access_logs WHERE  
timestamp > '2025-06-01';
```

```

7ad9c928729f :) CREATE TABLE yefremov_access_logs (
    event_date Date,
    user_id UInt32,
    timestamp DateTime,
    action String COMMENT 'Yefremov Aleksey action'
) ENGINE = MergeTree()
ORDER BY (event_date, user_id);

INSERT INTO yefremov_access_logs VALUES
('2025-06-01', 101, '2025-06-01 10:00:00', 'Yefremov login'),
('2025-06-02', 102, '2025-06-02 11:00:00', 'Aleksey view');

SELECT count()
FROM yefremov_access_logs
WHERE event_date ≥ '2025-06-01' AND event_date < '2025-07-01'
AND user_id IN (101, 102);

EXPLAIN ESTIMATE SELECT * FROM yefremov_access_logs WHERE timestamp > '2025-06-01';

CREATE TABLE yefremov_access_logs
(
    `event_date` Date,
    `user_id` UInt32,
    `timestamp` DateTime,
    `action` String COMMENT 'Yefremov Aleksey action'
)
ENGINE = MergeTree
ORDER BY (event_date, user_id)

Query id: 391e5bec-65c2-471e-9500-c29f6883f260

Ok.

0 rows in set. Elapsed: 0.014 sec.

INSERT INTO yefremov_access_logs FORMAT Values

Query id: 964a9e50-29e3-4d49-a641-9f9c636c1593

Ok.

2 rows in set. Elapsed: 0.009 sec.

SELECT count()
FROM yefremov_access_logs
WHERE (event_date ≥ '2025-06-01') AND (event_date < '2025-07-01') AND (user_id IN (101, 102))

Query id: 5043d3b1-6403-4fff-a622-ee340d90851b

1. 

|         |
|---------|
| count() |
| 2       |



1 row in set. Elapsed: 0.006 sec.

EXPLAIN ESTIMATE
SELECT *
FROM yefremov_access_logs
WHERE timestamp > '2025-06-01'

Query id: 029d22b2-5324-47ed-a52f-4919248ed88a

1. 

| database    | table                | parts | rows | marks |
|-------------|----------------------|-------|------|-------|
| yefremov_db | yefremov_access_logs | 1     | 2    | 1     |


```

Урок 7: Вторичные индексы в ClickHouse

Задание: Добавить вторичный индекс (skip-index), протестировать запросы.

Код:

```
CREATE TABLE yefremov_logs (  
    timestamp DateTime,  
    user_id UInt32,  
    action String,  
    INDEX yefremov_action_idx (action) TYPE minmax GRANULARITY 1  
) ENGINE = MergeTree() ORDER BY timestamp;
```

```
INSERT INTO yefremov_logs VALUES  
( '2025-06-01 10:00:00', 101, 'Yefremov action'),  
( '2025-06-02 11:00:00', 102, 'Aleksey action');
```

```
SELECT * FROM yefremov_logs WHERE action = 'Yefremov action';
```

```
EXPLAIN INDEXES=1 SELECT * FROM yefremov_logs WHERE action =  
'Yefremov action';
```

```

7ad9c928729f :) CREATE TABLE yefremov_logs (
    timestamp DateTime,
    user_id UInt32,
    action String,
    INDEX yefremov_action_idx (action) TYPE minmax GRANULARITY 1
) ENGINE = MergeTree() ORDER BY timestamp;

INSERT INTO yefremov_logs VALUES
('2025-06-01 10:00:00', 101, 'Yefremov action'),
('2025-06-02 11:00:00', 102, 'Aleksey action');

SELECT * FROM yefremov_logs WHERE action = 'Yefremov action';

EXPLAIN INDEXES=1 SELECT * FROM yefremov_logs WHERE action = 'Yefremov action';

CREATE TABLE yefremov_logs
(
    `timestamp` DateTime,
    `user_id` UInt32,
    `action` String,
    INDEX yefremov_action_idx action TYPE minmax GRANULARITY 1
)
ENGINE = MergeTree
ORDER BY timestamp

Query id: e039531f-684a-4396-9365-4e445d405100

Ok.

0 rows in set. Elapsed: 0.016 sec.

INSERT INTO yefremov_logs FORMAT Values

Query id: 293b3d20-cc6e-47f3-8c72-603376249daf

Ok.

2 rows in set. Elapsed: 0.008 sec.

SELECT *
FROM yefremov_logs
WHERE action = 'Yefremov action'

Query id: a1ef4f94-c9cf-471e-8730-f7cc1cfd9f31

1. | timestamp | user_id | action |
   |-----|-----|-----|
   | 2025-06-01 10:00:00 | 101 | Yefremov action |

1 row in set. Elapsed: 0.005 sec.

EXPLAIN INDEXES = 1
SELECT *
FROM yefremov_logs
WHERE action = 'Yefremov action'

Query id: 877a59eb-e0f7-447f-8760-4a2e5d3e5c64

Elapsed: 0.004 sec.

```

Урок 8: Лучшие практики оптимизации

Задание: Избегать функций в WHERE, использовать префиксы ключа.

Код:

```

SELECT count() FROM yefremov_access_logs
WHERE event_date BETWEEN '2025-06-01' AND '2025-06-30';

```

```
SELECT count() FROM yefremov_access_logs  
WHERE toYear(event_date) = 2025;
```

```
EXPLAIN SELECT count() FROM yefremov_access_logs WHERE  
toYear(event_date) = 2025;
```

```
7ad9c928729f :) SELECT count() FROM yefremov_access_logs  
WHERE event_date BETWEEN '2025-06-01' AND '2025-06-30';
```

```
SELECT count() FROM yefremov_access_logs  
WHERE toYear(event_date) = 2025;
```

```
EXPLAIN SELECT count() FROM yefremov_access_logs WHERE toYear(event_date) = 2025;
```

```
SELECT count()  
FROM yefremov_access_logs  
WHERE (event_date ≥ '2025-06-01') AND (event_date ≤ '2025-06-30')
```

Query id: 3660df24-caaa-48d8-8cae-6063e24f78ef

```
1. count()  
   2
```

1 row in set. Elapsed: 0.007 sec.

```
SELECT count()  
FROM yefremov_access_logs  
WHERE toYear(event_date) = 2025
```

Query id: befda3b9-54a2-40be-b65d-67f883947d99

```
1. count()  
   2
```

1 row in set. Elapsed: 0.006 sec.

```
EXPLAIN  
SELECT count()  
FROM yefremov_access_logs  
WHERE toYear(event_date) = 2025
```

Query id: 5ba6b375-2f38-4145-8729-05238a93fcf3

```
1. explain  
2. Expression ((Project names + Projection))  
3. Aggregating  
   ReadFromPreparedSource (_exact_count_projection)
```

3 rows in set. Elapsed: 0.004 sec.

7ad9c928729f :) █

Урок 9: Материализованные представления и проекции

Задание: Создать materialized view для подсчётов.

Код:

```
DROP VIEW IF EXISTS yefremov_mv_count;  
CREATE MATERIALIZED VIEW yefremov_mv_count
```

```
ENGINE = AggregatingMergeTree()
ORDER BY direction_code
AS SELECT direction_code, count() AS cnt
FROM yefremov_dissertations
GROUP BY direction_code;

INSERT INTO yefremov_dissertations (direction_code, author_id,
title, type, defense_date, organization, diploma_number) VALUES
('05.01', 1, 'Разработка сетей 5G', 'кандидатская', '2024-06-15',
'MГУ', 'DIP-001'),
('05.01', 2, 'Оптимизация телекоммуникаций', 'докторская', '2023-
09-10', 'СПбГУ', 'DIP-002'),
('01.01', 3, 'Алгоритмы машинного обучения', 'кандидатская',
'2025-01-05', 'ВШЭ', 'DIP-003'),
('05.01', 5, 'Системы связи', 'кандидатская', '2025-11-01',
'МТУСИ', 'DIP-005'),
('05.01', 9, 'Беспроводные сети', 'кандидатская', '2024-10-10',
'МТУСИ', 'DIP-009'),
('03.01', 6, 'Синтез органических соединений', 'кандидатская',
'2024-05-18', 'СПбГУ', 'DIP-006');

SELECT * FROM yefremov_mv_count
ORDER BY cnt DESC;

REFRESH MATERIALIZED VIEW yefremov_mv_count;
```

```

7ad9c928729f :) DROP VIEW IF EXISTS yefremov_mv_count;
CREATE MATERIALIZED VIEW yefremov_mv_count
ENGINE = AggregatingMergeTree()
ORDER BY direction_code
AS SELECT direction_code, count() AS cnt
FROM yefremov_dissertations
GROUP BY direction_code;

INSERT INTO yefremov_dissertations (direction_code, author_id, title, type, defense_date, organization, diploma_number) VALUES
('05.01', 1, 'Разработка сетей 5G', 'кандидатская', '2024-06-15', 'МГУ', 'DIP-001'),
('05.01', 2, 'Оптимизация телекоммуникаций', 'докторская', '2023-09-10', 'СПбГУ', 'DIP-002'),
('01.01', 3, 'Алгоритмы машинного обучения', 'кандидатская', '2025-01-05', 'ВШЭ', 'DIP-003'),
('05.01', 5, 'Системы связи', 'кандидатская', '2025-11-01', 'МТУСИ', 'DIP-005'),
('05.01', 9, 'Беспроводные сети', 'кандидатская', '2024-10-10', 'МТУСИ', 'DIP-009'),
('03.01', 6, 'Синтез органических соединений', 'кандидатская', '2024-05-18', 'СПбГУ', 'DIP-006');

SELECT * FROM yefremov_mv_count
ORDER BY cnt DESC;

REFRESH MATERIALIZED VIEW yefremov_mv_count;

DROP VIEW IF EXISTS yefremov_mv_count

Query id: df11e9cf-8871-451d-8aa9-48bf320e0a09

Ok.

0 rows in set. Elapsed: 0.003 sec.

CREATE MATERIALIZED VIEW yefremov_mv_count
ENGINE = AggregatingMergeTree
ORDER BY direction_code
AS SELECT
    direction_code,
    count() AS cnt
FROM yefremov_dissertations
GROUP BY direction_code

Query id: 655c282b-a594-44ec-9e49-92201235e4e1

Ok.

0 rows in set. Elapsed: 0.036 sec.

INSERT INTO yefremov_dissertations (direction_code, author_id, title, type, defense_date, organization, diploma_number) FORMAT Values

Query id: a48ee629-8a9e-49dc-815e-4f88eaada29a

Ok.

6 rows in set. Elapsed: 0.014 sec.

SELECT *
FROM yefremov_mv_count
ORDER BY cnt DESC

Query id: c0c7b074-6d28-4dd8-acb6-2b7143f7cb34



|    | direction_code | cnt |
|----|----------------|-----|
| 1. | 05.01          | 4   |
| 2. | 01.01          | 1   |
| 3. | 03.01          | 1   |


```

Урок 10: Настройка кластера и settings

Задание: Изменить settings, протестировать кластер

Код:

```

USE yefremov_db;

SELECT '=== Текущие настройки до изменения ===';

SELECT name, value
FROM system.settings
WHERE name IN ('max_threads', 'max_memory_usage',
'join_algorithm');

SET max_threads = 2;

SET max_memory_usage = 300000000;    -- ~300 МБ

SET join_algorithm = 'grace_hash';

SELECT '=== Настройки после изменения (Yefremov Aleksey) ===';

```

```
SHOW SETTINGS LIKE 'max_thread%';
SHOW SETTINGS LIKE 'max_memory%';
SHOW SETTINGS LIKE 'join_algorithm';
CREATE OR REPLACE TABLE yefremov_settings_test AS
SELECT number AS id, 'Yefremov Aleksey test ' || toString(number)
AS name
FROM numbers(1000000);
SELECT '=== Запрос с ограниченными настройками ===';
SELECT count() FROM yefremov_settings_test WHERE name LIKE
'%Yefremov%';
SELECT '=== Информация о кластере ===';
SELECT cluster, host_name FROM system.clusters LIMIT 5;
```

```

7ad9c928729f :) USE yefremov_db;
SELECT '=== Текущие настройки до изменения ===';
SELECT name, value
FROM system.settings
WHERE name IN ('max_threads', 'max_memory_usage', 'join_algorithm');
SET max_threads = 2;
SET max_memory_usage = 300000000; -- ~300 MB
SET join_algorithm = 'grace_hash';
SELECT '=== Настройки после изменения (Yefremov Aleksey) ===';
SHOW SETTINGS LIKE 'max_thread%';
SHOW SETTINGS LIKE 'max_memory%';
SHOW SETTINGS LIKE 'join_algorithm';
CREATE OR REPLACE TABLE yefremov_settings_test AS
SELECT number AS id, 'Yefremov Aleksey test ' || toString(number) AS name
FROM numbers(1000000);
SELECT '=== Запрос с ограниченными настройками ===';
SELECT count() FROM yefremov_settings_test WHERE name LIKE '%Yefremov%';
SELECT '=== Информация о кластере ===';
SELECT cluster, host_name FROM system.clusters LIMIT 5;

```

USE yefremov_db

Query id: 6b94ef1c-a00c-4667-9690-bd094586394d

Ok.

0 rows in set. Elapsed: 0.003 sec.

```
SELECT '=== Текущие настройки до изменения ==='
```

Query id: 1f18a180-1764-49f7-ad34-e4b4ff33efb7

```

1.  '=== Текущие настройки до изменения ==='
    '=== Текущие настройки до изменения ==='

```

1 row in set. Elapsed: 0.003 sec.

```

SELECT
  name,
  value
FROM system.settings
WHERE name IN ('max_threads', 'max_memory_usage', 'join_algorithm')

```

Query id: 0c7d1e2b-0795-4731-9f04-23c4da35edbb

	name	value
1.	max_threads	'auto(16)'
2.	join_algorithm	direct,parallel_hash,hash
3.	max_memory_usage	0

3 rows in set. Elapsed: 0.007 sec. Processed 1.43 thousand rows, 519.08 KB (193.86 thousand rows/s., 70.22 MB/s.)
Peak memory usage: 94.74 KiB.

```
SET max_threads = 2

Query id: 57ae988e-de3c-4cfe-96cd-6567f638f689

Ok.

0 rows in set. Elapsed: 0.003 sec.

SET max_memory_usage = 300000000

Query id: 515ebb85-735f-4d45-b2bb-ffd7e7bcd21

Ok.

0 rows in set. Elapsed: 0.003 sec.

SET join_algorithm = 'grace_hash'

Query id: 5580a497-428a-460d-a494-5c726ba8b991

Ok.

0 rows in set. Elapsed: 0.002 sec.

SELECT '=== Настройки после изменения (Yefremov Aleksey) ==='

Query id: 734b5c8d-9cea-4681-9e63-246fd3f21808

1.  ['=== Настройки после изменения (Yefremov Aleksey) ==='
    '=== Настройки после изменения (Yefremov Aleksey) ===']

1 row in set. Elapsed: 0.002 sec.

SHOW SETTINGS LIKE 'max_thread%'

Query id: 14b3fd8c-5fd0-4327-b323-7454728e5358

1.  | name | type | value |
2.  | max_threads | MaxThreads | 2 |
3.  | max_threads_for_annoy_index_creation | UInt64 | 4 |
    | max_threads_for_indexes | UInt64 | 0 |

3 rows in set. Elapsed: 0.008 sec. Processed 1.43 thousand rows, 519.07 KB (187.43 thousand rows/s., 67.89 MB/s.)
Peak memory usage: 110.39 KiB.

SHOW SETTINGS LIKE 'max_memory%'

Query id: 2da498d9-720c-4b0c-90ab-70a5ff2639c1

1.  | name | type | value |
2.  | max_memory_usage | UInt64 | 300000000 |
3.  | max_memory_usage_for_all_queries | UInt64 | 0 |
    | max_memory_usage_for_user | UInt64 | 0 |

3 rows in set. Elapsed: 0.007 sec. Processed 1.43 thousand rows, 519.07 KB (199.23 thousand rows/s., 72.17 MB/s.)
Peak memory usage: 110.36 KiB.

SHOW SETTINGS LIKE 'join_algorithm'
```

```
SHOW SETTINGS LIKE 'join_algorithm'
```

Query id: cd641e6e-4555-421a-b8bd-0b91d7e447b6

```
1. | name | type | value |
   |-----|-----|-----|
   | join_algorithm | JoinAlgorithm | grace_hash |
```

1 row in set. Elapsed: 0.007 sec. Processed 1.43 thousand rows, 519.07 KB (195.92 thousand rows/s., 70.97 MB/s.)
Peak memory usage: 110.39 KiB.

```
CREATE OR REPLACE TABLE yefremov_settings_test
AS SELECT
    number AS id,
    concat('Yefremov Aleksey test ', toString(number)) AS name
FROM numbers(1000000)
```

Query id: 02bbda60-d0b7-45e8-9615-af4e602fb16d

Ok.

1000000 rows in set. Elapsed: 0.143 sec. Processed 1.00 million rows, 8.00 MB (6.99 million rows/s., 55.92 MB/s.)
Peak memory usage: 75.74 MiB.

```
SELECT '=== Запрос с ограниченными настройками ==='
```

Query id: bf10a573-8b55-4a0e-8901-e53d3d2c0a01

```
1. | '=== Запрос с ограниченными настройками ===' |
   |-----|
   |=== Запрос с ограниченными настройками ===|
```

1 row in set. Elapsed: 0.002 sec.

```
SELECT count()
FROM yefremov_settings_test
WHERE name LIKE '%Yefremov%'
```

Query id: 0e330f5c-403c-4074-8522-e392597572b7

```
1. | count() |
   |-----|
   | 1000000 |
```

1 row in set. Elapsed: 0.033 sec. Processed 1.00 million rows, 35.89 MB (30.23 million rows/s., 1.08 GB/s.)
Peak memory usage: 1.43 MiB.

```
SELECT '=== Информация о кластере ==='
```

Query id: cc4f44ee-1b7e-45fb-a080-8a53224a7680

```
1. | '=== Информация о кластере ===' |
   |-----|
   |=== Информация о кластере ===|
```

1 row in set. Elapsed: 0.002 sec.

```
SELECT
    cluster,
    host_name
FROM system.clusters
LIMIT 5
```

Query id: 5947fcd1-e21b-4c61-b9bf-fb3a1a601e99

```
1. | cluster | host_name |
   |-----|-----|
   | default | localhost |
```

1 row in set. Elapsed: 0.003 sec.